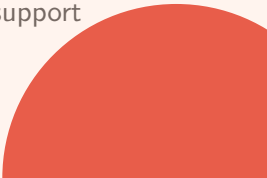


Modern AutoCrit + MetricSpace

An agentic clinical data-science tool for comparing eligibility criteria

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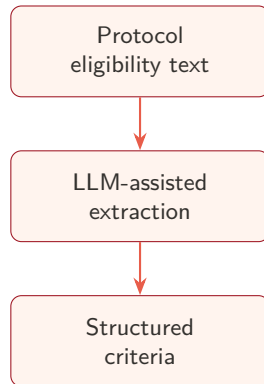
Project proposal: clinical protocols, agentic AI, and validated decision support



Core idea

Use an LLM to convert free-text clinical-trial eligibility criteria into a more structured representation.

- ▶ Makes protocol prose easier to process computationally.
- ▶ Establishes the protocol-to-structure workflow.
- ▶ Motivates automated cleaning and normalization.



From proof of concept to usable extraction tool

AutoCriteria demonstrated LLM-based protocol structuring. Modern AutoCrit makes that workflow inspectable, reviewable, and reusable for downstream clinical data science.

Engineering changes

- ▶ Modular browser application instead of one long script.
- ▶ Input by NCT ID, pasted text, or full protocol PDF.
- ▶ ClinicalTrials.gov API v2 retrieval.
- ▶ Removed LangChain/LangGraph dependencies in favor of direct SDK calls and explicit workflow code.
- ▶ Fewer hidden abstractions make structured output, failures, and debugging easier to trace.

More atomic, computable output

- ▶ Separates compound sentences into independently comparable criteria.
- ▶ Domain, entity, comparator, value, unit, and interval.
- ▶ Negation, timing, repetition, qualifiers, and AND/OR logic.
- ▶ Preserves source evidence while normalizing terminology.
- ▶ Human review plus versioned JSON/CSV and Excel export.

Source sentence

“Exclude blood pressure $> 145/95$ on two occasions within three months or a history of myocardial infarction.”

Atomic criterion 1

- ▶ Entity: blood pressure
- ▶ Comparator/value: $> 145/95$
- ▶ Repetition: two occasions
- ▶ Time: within three months
- ▶ Logic: OR; type: exclusion

Atomic criterion 2

- ▶ Entity: myocardial infarction
- ▶ Value: present
- ▶ Temporal construct: history of
- ▶ Logic: OR; type: exclusion

Cleaner comparison units $\rightarrow$ clearer labels $\rightarrow$ better embedding supervision

- ▶ Conventional embeddings mostly reward similar wording.
- ▶ Small textual changes can create large clinical differences.
- ▶ Cosine similarity cannot express directional relationships such as “A is more restrictive than B.”
- ▶ Complex sentences add negation, exceptions, severity, timing, and repeated measurements.

Canonical failure

age ≥ 18

age < 18

Nearly identical text
Disjoint populations

Tool need

Clinical data scientists need to find comparable criteria, see why they differ, detect contradictions, and route uncertain cases for review.

Oncology: semantic complexity

- ▶ Disease stage and histology
- ▶ Biomarkers and measurable disease
- ▶ ECOG/performance status
- ▶ Prior lines of therapy and washout
- ▶ Organ-function laboratories
- ▶ Exceptions and nested logic

Cardiovascular: mixed structure

- ▶ Blood pressure and ejection fraction
- ▶ Prior MI, stroke, or revascularization
- ▶ Medication contraindications
- ▶ Event timing and stability
- ▶ “Clinically significant” disease
- ▶ Repeated measurements

Practical use: cross-trial comparison, protocol harmonization, screening-logic review, and cohort-definition reuse

1. Where do conventional embeddings fail?

Across oncology and cardiovascular protocols, which operators, thresholds, negations, time windows, exceptions, and compound statements create clinically misleading similarity?

2. Can reviewed clinical supervision improve the representation?

Does fine-tuning on human-validated criterion pairs improve retrieval and relationship prediction over lexical, general, biomedical, and LLM-only baselines?

3. Does the agent make the tool safer and more useful?

Can tool selection, consistency checks, uncertainty estimates, and human escalation improve accuracy and calibration while reducing unnecessary review?

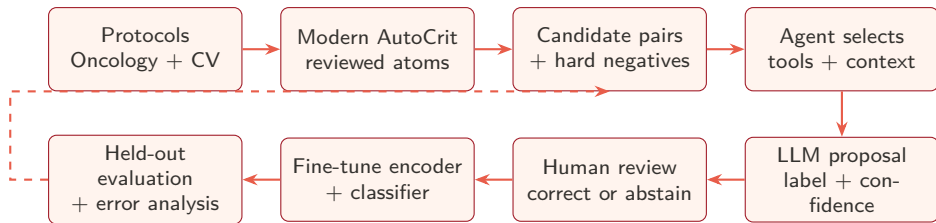
Pair relationship labels

- ▶ Equivalent
- ▶ A more restrictive than B
- ▶ B more restrictive than A
- ▶ Partially overlapping
- ▶ Contradictory/disjoint
- ▶ Related but not comparable
- ▶ Unrelated
- ▶ Ambiguous/insufficient information

Two outputs, not one opaque score

1. **Dense representation**: retrieve clinically comparable criteria.
2. **Relationship classifier**: predict equivalence, direction, overlap, contradiction, or abstention.

The interface should return the closest criteria, an interpretable relationship, supporting differences, and uncertainty.



The agent powers the user workflow

It retrieves protocol context and selects the appropriate comparison path. Explicit operators, intervals, and Boolean relations are evaluated with structured comparison tools; context-dependent clinical language is interpreted with embedding and LLM tools. The agent reconciles their evidence, records uncertainty, and escalates unresolved cases. Accepted corrections improve future versions of the tool.

1. Computable relation

age ≥ 18 versus age < 18

- ▶ Parse operators and intervals
- ▶ Calculate overlap or disjointness
- ▶ Return an exact, inspectable result

2. Contextual relation

Cardiovascular history versus controlled hypertension

- ▶ Retrieve clinical context
- ▶ Use dense retrieval and LLM judgment
- ▶ Report rationale and uncertainty

3. Agent decision

- ▶ Select one or both paths
- ▶ Check for conflicting evidence
- ▶ Accept, abstain, or request review

Human review is reserved for uncertain or conflicting cases—not every computable comparison.

Baselines and training

- ▶ TF-IDF and frozen general/biomedical encoders
- ▶ Automated structured comparison where exact logic is available
- ▶ LLM judgment without fine-tuning
- ▶ Contrastive fine-tuning with positives, ordinary negatives, and hard negatives
- ▶ Directional classification head

Population comparisons, if feasible, use aggregate publication characteristics—not unavailable individual patient data or unsupported simulation.

Evaluation

- ▶ Protocol-level train/validation/test splits
- ▶ Retrieval recall@ k and ranking quality
- ▶ Macro-F1 and class-specific errors
- ▶ Calibration and input-order consistency
- ▶ Reviewer agreement, correction rate, and abstention quality
- ▶ Ablations of human corrections, critic, and structured checks

Build and validate

1. Select comparable oncology and cardiovascular trials.
2. Run and human-review Modern AutoCrit outputs.
3. Freeze a versioned criterion dataset with provenance.
4. Build diverse criterion pairs—not age alone.
5. Run one frozen embedding baseline and identify failures.
6. Train, calibrate, and test the agentic comparison pipeline.

User-facing prototype

- ▶ Search or upload protocol criteria
- ▶ Retrieve comparable criteria across trials
- ▶ Show exact structured differences
- ▶ Classify clinical relationship
- ▶ Display confidence and rationale
- ▶ Escalate ambiguous cases to a reviewer
- ▶ Export reviewed comparisons for analysis

Modern AutoCrit structures what a protocol says.

MetricSpace learns how criteria relate clinically.

Protocol-centered Agentic Human-reviewed Statistically validated

Proposed deliverable

A working clinical data-science application that extracts reviewed criteria, retrieves comparable rules across trials, explains their relationship, quantifies uncertainty, and preserves an audit trail—supported by a validated dense representation and agentic quality-control workflow.

Starting references: Datta et al., *AutoCriteria* (2023); Kury et al., *Chia* (2020); Modern AutoCrit repository.